1. If $y = |\cos x| + |\sin x|$ then $\frac{dy}{dx}$ at $x = \frac{2\pi}{3}$ is

(a) $\frac{1-\sqrt{3}}{2}$ (b) 0

(c) $\frac{1}{2}(\sqrt{3}-1)$ (d) None of these

2. If $y = x^2 e^x$, then value of y_n is

(a) $\{x^2 - 2nx + n(n-1)\}e^x$ (b) $\{x^2 + 2nx + n(n-1)\}e^x$

(c) $\{x^2 + 2nx - n(n-1)\}e^x$ (d) None of these

3. If $y = 4x - 5$ is a tangent to the curve $y^2 = px^3 + q$ at (2,3) then :

(a) p = 2, q = -7 (b) p = -2, q = 7

(c) p = -2, q = -7 (d) p =2, q = 7

4. The function f(x) = $\left|\frac{x^2-2}{x^2-4}\right|$ has

(a) No point of local minima

(b) No point of local maxima

(c) Exactly one point of local minima

(d) Exactly one point of local maxima

5. The equation of the tangent to the curve f (x) = 1 + e^{-2x} where it cuts the line y = 2 is

(a) x + 2y = 2 (b) 2x + y = 2

(c) x - 2y = 1 (d) x - 2y + 2 = 0

6. The set of all values of a for which the function

$f(x) = \left(\frac{\sqrt{a+4}}{1-a} - 1\right)x^5 - 3x + \log 5$ decreases for all real x is

(a) $\left(-3, \frac{5-\sqrt{27}}{2}\right) \cup (2, \infty)$ (b) $\left[-4, \frac{3-\sqrt{21}}{2}\right] \cup (1, \infty)$

(c) $(-\infty, \infty)$ (d) $[1, \infty)$

7. $f(x) = \sin^p x.\cos^q x \left(p, q > 0, 0 < x < \frac{\pi}{2}\right)$ has point of maximum at

(a) $x = \tan^{-1}\left(\frac{\sqrt{p}}{q}\right)$ (b) $x = \tan^{-1}\left(\frac{\sqrt{q}}{p}\right)$

(c) No such point exist (d) None of these

8. If the function f(x) = $|x^2 + a|x| + b|$ has exactly three points of non-differentiability, then which of the following may hold

(a) b = 0, a < 0 (b) b < 0, a $\in$ R

(c) b > 0, a $\in$ R (d) b < 0, a $\in$ R^-

9. If $x^y = e^{x-y}$ then $\frac{dy}{dx}$ is equal to -

(a) $(1 + \log x)^{-1}$ (b) $(1 + \log x)^{-2}$

(c) $(\log x)(1 + \log x)^{-2}$ (d) None of these

10. If $u = f(x^2)$, $v = g(x^3)$; $f'(x) = \sin x$, $g'(x) = \cos x$, then $\frac{du}{dv}$ =

(a) $\frac{2\sin(x^2)}{3x(\cos x^3)}$ (b) $\frac{2}{3x}$ tan x

(c) $\frac{2}{3x}\frac{\sin^2 x}{\cos^3 x}$ (d) None

11. The equation of tangent to the graph of the function f(x) = $|x^2 - |x||$ at the point with abscissa x = -2 is

(a) 3x + y + 4 = 0 (b) 5x - y - 12 = 0

(c) 5x + y + 8 = 0 (d) 3x + y - 4 = 0

12. The value of 'a' for which the equation

$x^3 - 3x + a = 0$ has two different roots in [0, 1] is given by -

(a) -1 (b) 2 (c) 1 (d) None of these

13. The min. distance between y - x = 1 and $y^2 = x$ is -

(a) $\frac{3}{4\sqrt{2}}$ (b) $\frac{3}{2\sqrt{2}}$

(c) $\frac{1}{4\sqrt{2}}$ (d) $\frac{1}{2\sqrt{2}}$

14. $f(x) = e^{2x} - (a + 1)e^x + 2x$ is monotonic increasing for all $x \in R$ is

(a) (3, 4) (b) $(-\infty, 0)$ (c) $(-\infty, 3]$ (d) $(3, \infty)$

15. Function f(x) = $\frac{a\sin x + b\cos x}{c\sin x + d\cos x}$ is monotonic decreasing if -

(a) ad - bc < 0 (b) ad - bc > 0

(c) ab - cd < 0 (d) ab - cd > 0

16. In its domain, f(x) = $\frac{\sin^{-1} x}{\cot^{-1} x}$ is -

(a) A increasing function

(b) A strictly increasing function

(c) A decreasing function

(d) A strictly decreasing function

17. Let $f(x) = e^x \cos x$ and slope of the curve y = f(x) is maximum at x = a then a equals

(a) 0 (b) $\frac{\pi}{2}$ (c) $\frac{3\pi}{2}$ (d) None of these

18. Let f(x) = $\sin\left(\frac{\{x\}}{a}\right) + \cos\left(\frac{\{x\}}{a}\right)$, {.} denotes the fraction part. Then the set of values of a, for which f(x) can attain its maximum value is -

(a) $(0, 4/\pi)$ (b) $(\pi/4, \infty)$

(c) $(0, \infty)$ (d) None

19. If x = - 1 and x = 2 are extreme points of the function $y = a \log x + bx^2 + x$, then

(a) a = 2, b = 1/2 (b) a = 2, b = - ½

(c) a = -2, b = ½ (d) a = - 2, b = - ½

20. If $y = \sin^{-1}(\cos x)$, $x \in \left(0, \frac{\pi}{2}\right)$ then $\frac{dy}{dx}$ is -

(a) -1 (b) sin x (c) $\sin^2 x$ (d) None of these